MAST-U OpsSplit: Explicit and Implicit Operator Splits

JOREK model 303, axisymmetric $n = 0$ — 80/20 perturbation ensemble — interim results

Result. Carrying the complete frozen model303 tangent in an implicit step lowers held-out error 9–60× relative to NODE and the explicit continuity OpsSplit. Giving that frozen physics the true $D_\perp$ lowers out-of-distribution error a further 6–29×. The gain comes from the frozen native physics: the untrained physics-only model matches the trained arms wherever both are complete. Unfinished runs are marked pending and are never averaged.

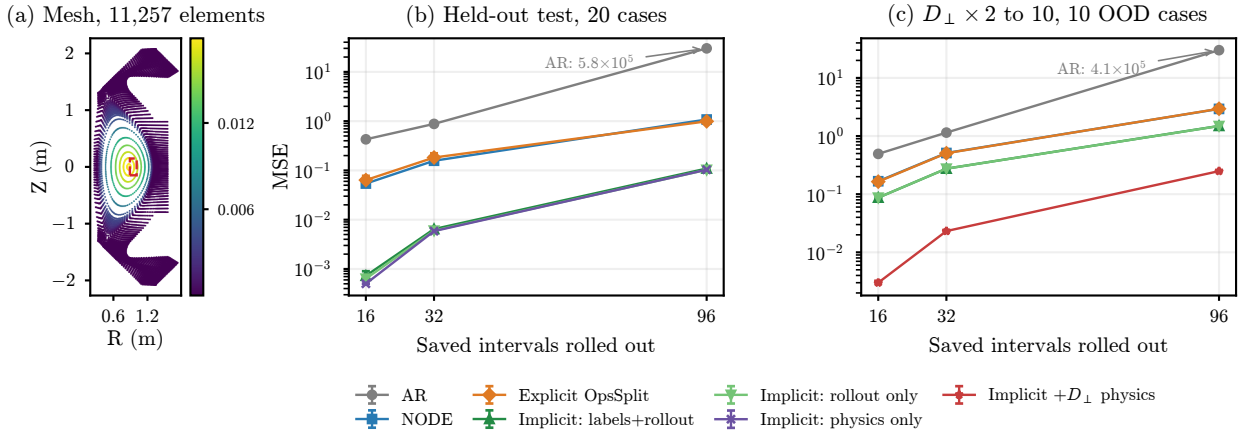


Figure 1: (a) Native bicubic mesh coloured by the training-mean pressure; dashed box: range of perturbation centres. (b, c) Normalized volume MSE over all seven fields; seed means with seed min–max bars, one Euler substep. Off-scale AR points are labelled. OOD networks receive the training conditioning value 1.

1 Case, data and split

Unmodified JOREK model303 evolves one axisymmetric MAST-U equilibrium with seven fields $q = (\psi, u, z_j, w, \rho, T, V_\parallel)$. The state is the native bicubic coefficient vector on one fixed mesh: 11,257 elements, 46,203 DOFs per field and 323,421 unknowns. Each of the 100 trajectories multiplies ρ and T by $1 + A_f S(R, Z, \psi_N)$. S is a compact Gaussian bump (centre $R = 0.89$–1.01 m, $|Z| \le 0.15$ m; widths 0.07–0.10 by 0.12–0.20 m), tapered to zero between $\psi_N = 0.55$ and 0.75. The amplitudes are $|A_f| = 0.01$–0.05, with a random sign per field. Runs are saved every 14.6 ns for 96 intervals (1.41 μs).

The split was frozen before any simulation as 60/20/20. On 6 November validation joined training, giving 80/0/20; every model uses its final update. The OOD set reruns 10 held-out test initial states with $D_\perp$ multiplied by 2–10. It never trains or normalizes anything.

2 Shared protocol

Network. Every family uses one L-SplineFNO N_θ (16×32 latent grid, 4 modes, width 8, two layers; 9,673 parameters). The consistent-mass projection P maps its point outputs to native coefficients. A FiLM input carries a coefficient multiplier that is 1 in all training.

Guard $\mathcal{G}$. After every step of every family, ψ and u are recomputed from the predicted z_j and w through their native weak (elliptic) definitions, with fixed boundary values. A subdivided-Bernstein limiter then keeps ρ and T positive on every cell by blending toward the training mean. Negative proposals add a penalty (weight 10^{-3}) to the loss.

Training. Seeds 17/18/19; within a seed, every family starts from identical weights and sees the same sampled windows. Each run takes 2,000 Adam updates (learning rate 10^{-3}, gradient clip 2, batch one), using training-only normalization. Each update rolls out four saved intervals and averages the volume-weighted normalized MSE over them.

Evaluation. Each test trajectory runs autonomously from frame 0 for 16/32/96 intervals (0.23/0.47/1.41 μs), scored by normalized volume MSE over all seven fields. A horizon counts only if every trajectory passes the positivity, constraint and boundary audits. All families except AR also run 2 and 4 substeps.

3 How each method is conducted

Here h is one saved interval and s the number of Euler substeps.

AR. $q^{n+1} = \mathcal{G}\big(PN_\theta(q^n)\big)$: the network predicts the next state directly.

Euler NODE. $q \leftarrow \mathcal{G}\big(q + \frac{h}{s} PN_\theta(q)\big)$, repeated s times.

Table 1: Normalized volume MSE, mean over completed seeds, one substep; "Seeds": completed/expected; "run.": running; "pend.": not started; "n.i.": not interpretable (Section 4); "–": not evaluated by design. Train: summed wall time of the optimizer updates per seed, mean over finished seeds, without setup or evaluation; explicit families on 4 CPUs (2 threads), implicit arms on 8 (4 threads). $D_\perp$ rows reuse the trained checkpoints. Labels + rollout also needs a one-off 2,000-state native label export (8.8 h summed, parallel).

		Held-out test (20 cases)				$D_\perp$ OOD (10 cases)			
Method	Train (h)	Seeds	16	32	96	Seeds	16	32	96
AR	2.5	1/1	0.43	0.88	5.8×10^{5}	1/1	0.49	1.15	4.1×10^{5}
Euler NODE	2.5	1/1	0.054	0.16	1.08	1/1	0.17	0.51	2.93
Explicit OpsSplit, learned ∇	4.0	3/3	0.064	0.18	0.99	1/1	0.16	0.51	2.95
Explicit OpsSplit, native ∇	3.0	3/3	0.064	0.18	0.99	–	–	–	–
Explicit OpsSplit $+ D_\perp\nabla^2\rho$	5.6	1/1	0.052	0.15	1.04	1/1	n.i.	n.i.	n.i.
Implicit: labels + rollout	5.2	3/3	7.3×10^{-4}	6.5×10^{-3}	0.11	3/3	0.087	0.27	1.49
Implicit: rollout only	4.6	1/3	6.3×10^{-4}	6.0×10^{-3}	0.10	1/3	0.087	0.27	1.49
Implicit: physics only ($N=0$)	0	1/1	5.0×10^{-4}	5.9×10^{-3}	0.10	0/1	pend.	pend.	pend.
$+ D_\perp$ physics: labels + rollout	–	–	–	–	–	3/3	3.0×10^{-3}	0.023	0.25
$+ D_\perp$ physics: rollout only	–	–	–	–	–	1/3	2.9×10^{-3}	0.023	0.24
$+ D_\perp$ physics: physics only	–	–	–	–	–	1/1	2.6×10^{-3}	0.022	0.24

Explicit physical OpsSplit. NODE plus a frozen physical branch:

$$q \leftarrow \mathcal{G}\Big(q + \tfrac{h}{s}\big[PN_\theta(q) + PL_c(q)\big]\Big).$$

L_c is the exact linearization of model303's density and temperature advection–compression, $\rho_t = \mathbf{a}\cdot\nabla\rho + c\rho$ and $T_t = \mathbf{a}\cdot\nabla T + (\gamma-1)cT$, about the training-mean background. It acts on ρ and T only and adds no parameters. State gradients come from the network's learned spline synthesis; a gradient study retrains three seeds with exact native bicubic gradients. A seed-17 variant also adds the known density diffusion $D_\perp\nabla^2\rho$.

Implicit native-tangent OpsSplit. The full model303 volume right-hand side is split about a warm baseline state q^*: $F(q) = F(q^*)+L^*(q-q^*)+N(q)$. L^* is the complex-step tangent of the complete pointwise RHS for all seven fields, with transport and sources fixed. M^* is the native mass (temporal) operator at q^*. One step solves

$$(M^* - hL^*)\,q' = M^*q + h\big(F(q^*) - L^*q^* + N_\theta(q)\big)$$

and then applies $\mathcal{G}$, reusing one cached sparse LU per step size. Current and vorticity rows are their weak definitions, and DOFs that never change in training data are Dirichlet rows. JOREK's Mach-1 wall condition on $V_\parallel$ is reproduced by 870 data-identified algebraic rows, and 513 further wall rows are held. N_θ is the same backbone with 13 weak-form output channels (9,727 parameters). Its output layer starts at zero, so the untrained model is exactly the physics-only model. There are three arms:

- *Labels + rollout:* 2,000 updates on exact native remainder labels (2,000 training states), then the shared 2,000 rollout updates.
- *Rollout only:* the 2,000 rollout updates from the physics-only start.
- *Physics only:* $N = 0$, no training.

Coefficient physics for OOD. Only $D_\perp$ changes in the OOD set, and the native RHS is exactly affine in it. This variant therefore updates $F(q^*)$ and L^* for each case's true $D_\perp$, refactorizes, and applies the remaining $D_\perp$ term explicitly. The network and its conditioning value of 1 are unchanged. A native check reproduces the forcing and tangent changes to 4.3×10^{-16} and 7.4×10^{-16} relative error. Other families see $D_\perp$ only through FiLM, trained at 1, so their OOD input stays 1.

4 Interim results

Substeps change NODE and explicit OpsSplit by at most 2.4%, and exact native gradients change explicit OpsSplit by at most 0.28%. At 96 intervals, the implicit arms beat NODE in all 106 paired test comparisons (arm, seed and trajectory) and all 40 OOD comparisons. Their remaining test error is 84–91% current z_j. With $D_\perp$ withheld, their OOD error is 90–90% ρ and T; supplying the true $D_\perp$ removes most of it. The untrained physics-only arm matches both trained arms, and label pretraining lowers the normalized remainder loss only from 1 to 0.86–0.92: the 9,727-parameter remainder adds little, so a larger one is the next lever. The density-diffusion OOD run also fed $D_\perp$ into FiLM, so it is not interpretable.

5 Caveats

AR and NODE have one seed (18/19 cancelled); explicit OpsSplit seeds 18/19 come from the gradient study's matched arm. M^* and L^* are frozen at q^*; the small perturbations share one equilibrium and mesh, so mesh convergence and cross-equilibrium generalization are untested. With exact labels and more CPUs, the implicit family is not compute-matched.